Two-Seller Competition: Model and Extinction Questions

Progress note — transcription of handwritten notes

October 7, 2026

This note transcribes the three handwritten pages into English. The initial intuition and its subsequent qualification are presented in their original order.

1 Elements of the model

- Two symmetric providers, indexed by $j \in \{A, B\}$.
- Two products, indexed by $i \in \{1, 2\}$.
- Success probabilities: $0 < p_1 < p_2 < 1$.
- Costs: $c_1 < c_2$.
- Revenue: $R > c_2$.

Customer beliefs. The customer observes the state

$$s = (S_A, F_A, S_B, F_B),$$

where S_j and F_j count provider j's successes and failures. Her beliefs are

$$\theta_A \mid S_A, F_A \sim \text{Beta}(1 + S_A, 1 + F_A), \tag{1}$$
$$\theta_B \mid S_B, F_B \sim \text{Beta}(1 + S_B, 1 + F_B). \tag{2}$$

For the first model, we assume that *both sellers observe the full state s.*

Customer choice. The customer uses Thompson sampling (TS), drawing independently from the two belief distributions and choosing

$$y_t = \begin{cases} A, & \widetilde{\theta}_A > \widetilde{\theta}_B, \\ B, & \widetilde{\theta}_B > \widetilde{\theta}_A. \end{cases} \tag{3}$$

Thus the selection probabilities are

$$D_A(s) = \mathbb{P}(\widetilde{\theta}_A > \widetilde{\theta}_B \mid s), \qquad D_B(s) = 1 - D_A(s). \tag{4}$$

Ties have probability zero under the continuous Beta distributions.

2 Timing, state transitions, and strategies

Timing.

1. Everyone observes s.
2. The customer chooses a seller using Thompson sampling.
3. Only the selected provider chooses product i.
4. The customer observes the outcome, but not the product.
5. The selected provider's state is updated.

State transitions. Superscripts s and f denote success and failure, respectively:

$$\begin{aligned} T_A^s(s) &= (S_A + 1, F_A, S_B, F_B), \\ T_A^f(s) &= (S_A, F_A + 1, S_B, F_B), \\ T_B^s(s) &= (S_A, F_A, S_B + 1, F_B), \\ T_B^f(s) &= (S_A, F_A, S_B, F_B + 1). \end{aligned} \tag{5}$$

Strategies. The sellers' product choices, conditional on selection, are

$$\pi_A(s) \in \{1, 2\}, \qquad \pi_B(s) \in \{1, 2\}. \tag{6}$$

Here π_j denotes a product-choice strategy, as in the handwritten notes.

3 Bellman equation and equilibrium

Bellman equation. Fix seller B's strategy π_B. Let $V_A(s)$ be seller A's optimal discounted value before customer selection, and let $\gamma \in (0, 1)$ denote the discount factor. Then

$$\begin{aligned} V_A(s) = D_A(s) \max_{i \in \{1,2\}} \Big\{ R - c_i + \gamma \big[p_i V_A(T_A^s(s)) + (1 - p_i) V_A(T_A^f(s)) \big] \Big\} \\ + D_B(s) \gamma \Big[p_{\pi_B(s)} V_A(T_B^s(s)) + (1 - p_{\pi_B(s)}) V_A(T_B^f(s)) \Big]. \end{aligned} \tag{7}$$

With π_A fixed, seller B's Bellman equation is symmetric.

Equilibrium. A pair (π_A^*, π_B^*) is a Markov perfect equilibrium if, at every state s, $\pi_A^*(s)$ solves the maximization given π_B^*, and $\pi_B^*(s)$ solves the corresponding maximization given π_A^*. Equivalently,

$$\boxed{\pi_A^* \in \mathrm{BR}_A(\pi_B^*), \qquad \pi_B^* \in \mathrm{BR}_B(\pi_A^*).} \tag{8}$$

Product 2. Conditional on being selected, we obtain the same threshold. Define $\Delta c = c_2 - c_1$ and $\Delta p = p_2 - p_1$. For seller A,

$$\boxed{\text{Product 2 is optimal} \quad \Longleftrightarrow \quad \gamma \big[V_A(T_A^s(s)) - V_A(T_A^f(s)) \big] \geq \frac{\Delta c}{\Delta p}.} \tag{9}$$

At equality, both products are optimal.

4 Initial intuition and computational concerns

Initial intuition. Extinction should continue to hold because the customer will eventually have accumulated so much information that her beliefs will change very little. Thus, both sellers will eventually stop using product 2.

Problems and possible approaches.

- In the base model, the state space is infinite (the state keeps growing), which makes implementation computationally demanding.
- Perhaps it would be better to work with forgetting.
- If so, we could take advantage of the boundary.

5 Does extinction occur?

Not always, or more precisely, not at every state. Why? There are cases in which the difference in demand does not tend to zero as n grows.

Example. Consider the sequence of states

$$s_n = (0, n, 0, n), \qquad n \in \mathbb{N}_0. \tag{10}$$

In particular, the customer's draw for seller B satisfies $\widetilde{\theta}_B \sim \text{Beta}(1, n+1)$. Calculate seller A's demand after each possible outcome:

$$\text{Success:} \quad D_A(T_A^s(s_n)) = \mathbb{P}\big(\text{Beta}(2, n+1) > \text{Beta}(1, n+1)\big) = \frac{3n+4}{4n+6}, \tag{11}$$

$$\text{Failure:} \quad D_A(T_A^f(s_n)) = \mathbb{P}\big(\text{Beta}(1, n+2) > \text{Beta}(1, n+1)\big) = \frac{n+1}{2n+3}. \tag{12}$$

The Beta notation inside these probabilities represents independent random variables with the indicated distributions. Therefore,

$$\begin{aligned} D_A(T_A^s(s_n)) - D_A(T_A^f(s_n)) &= \frac{3n+4}{4n+6} - \frac{n+1}{2n+3} \\ &= \frac{n+2}{4n+6} = \frac{1+2/n}{4+6/n} \longrightarrow \frac{1}{4}, \end{aligned} \tag{13}$$

In this case, because the posteriors concentrate around zero, seller A may still benefit from using product 2. We cannot guarantee that the difference in demand is small at every state simply because the total number of observations is large.

Note. The calculation above concerns the demand difference which is the tool that we used in the base case to proof the extinction result. (9).

The thing with this example is that the probability of that state occurring is very small, and that is an important point. Maybe we need to study individual trajectories to understand what happens?

Computational problem. Because the state grows without bound, implementing the full problem is infeasible or computationally very demanding. We would need to truncate the problem.

What could we prove. Existence of equilibrium?